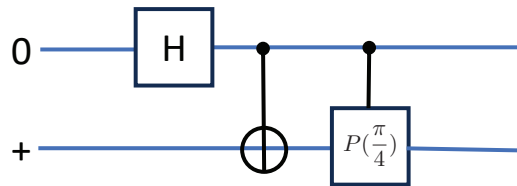


CSCI5244: Quantum Computation and Information

Homework Set 1

1. Given a state of a qubit $|\psi\rangle = a|0\rangle + b|1\rangle$, where $a, b \in \mathbb{C}$ and $|a|^2 + |b|^2 = 1$, and recall that states $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$.
 - (a). In 2-dimensional complex space with basis $\{|0\rangle, |1\rangle\}$, write down the matrix form of operator $A = |+\rangle\langle+| + \frac{1}{2}|-\rangle\langle-|$, and verify A is an observable (i.e., self-adjoint).
 - (b). If you measure the observable A on the state $|\psi\rangle$, what are the possible measurement outcomes and the associated probabilities?
2. Consider a state of two qubits $|\psi\rangle = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle - \frac{i}{\sqrt{2}}|11\rangle$.
 - (a). If you label the basis of 4-dimensional complex space in the following order: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, what are the vector forms of $|\psi\rangle$ and $\langle\psi|$?
 - (b). If you label the basis of 4-dimensional complex space in the following order: $|00\rangle, |10\rangle, |01\rangle, |11\rangle$, what are the vector forms of $|\psi\rangle$ and $\langle\psi|$?
 - (c). If you label the basis of 4-dimensional complex space in the following order: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, what is the matrix form of density operator $\rho = |\psi\rangle\langle\psi|$?
 - (d). Is $|\psi\rangle$ a separable state or entangled state?
3. Recall the quantum gate notation and diagrams introduced in lecture 2, and consider the following circuit with the input $|0\rangle|+\rangle$.



- (a). What is the output of the above circuit?
- (b). We measure the second qubit (denoted by the lower horizontal line in the circuit) on the computational basis $\{|0\rangle, |1\rangle\}$ at the end of computation. After the measurement, what are the possible states of the first qubit, and the associated probabilities?